

Notes: König, Storesletten, Song, and Zilibotti (Econometrica, 2022)

From Imitation to Innovation: Where Is All That Chinese R&D Going?

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1 Big picture

- **Question.** Where is the Chinese R&D boom going, and how much does firm-level misallocation distort the assignment of firms between imitation and innovation?
- **Setting.** The paper combines an endogenous-growth model with random interactions and firm-level distortions with Chinese manufacturing data for 2007–2012, and then compares the estimates to Taiwanese manufacturing data for 1999–2004.
- **Core challenge.** A lot of R&D spending does not automatically imply fast productivity growth. What matters is which firms innovate, which firms imitate, and how distortions affect that assignment.
- **Main conceptual contribution.** The paper embeds *innovation* and *imitation* in one common framework. Low-TFP firms may optimally imitate because they have more to gain from learning from others; high-TFP firms may optimally innovate because imitation opportunities are limited near the frontier.
- **Main empirical contribution.** Instead of taking the distribution of productivity as exogenous, the paper structurally estimates a model in which the TFP distribution is itself an equilibrium object shaped by firm choices, distortions, and random interactions.
- **Main empirical design.** Firms with positive R&D are treated as innovators and firms without R&D as imitators. The authors estimate the stationary equilibrium of the model by simulated method of moments, targeting the empirical relation between R&D, TFP, firm size, and subsequent TFP growth.
- **Main reduced-form facts.** In China, R&D is more common among high-TFP firms and among larger firms, but the size relation is steeper than the TFP relation. Conditional on TFP, R&D firms grow faster; among non-R&D firms, TFP growth is strongly decreasing in initial TFP, consistent with imitation-driven convergence.
- **Main structural result.** In the benchmark parsimonious model, the estimated economy generates an annual aggregate TFP growth rate of about 3.6%, close to the data even though aggregate growth is *not* directly targeted.
- **Main misallocation result.** Output wedges have large dynamic effects: they induce the wrong firms to innovate and the wrong firms to imitate. Lower misallocation raises long-run TFP growth and changes the whole shape of the TFP distribution.
- **Main comparative result.** Taiwan looks broadly similar to China, but the productivity of innovation and the speed of technology diffusion are both higher in Taiwan.
- **Main policy result.** Moderate broad-based R&D subsidies can raise growth, but very large subsidies can backfire because they push too many low-TFP firms away from high-return imitation and into low-return innovation.
- **Economic takeaway.** The paper is not really about “more R&D is always better.” It is about *who* should innovate, *who* should imitate, and how misallocation distorts that assignment in a way that slows growth.

2 Data and measurement (key objects)

2.1 Why China and Taiwan are useful laboratories

- The Chinese data come from the Annual Survey of Industries collected by the National Bureau of Statistics. The sample used for the main estimation is a balanced panel of manufacturing firms observed in both 2007 and 2012.
- The Taiwanese comparison sample is plant-level census data for 1999–2004. Taiwan is useful because it is culturally and geographically close to China, still heavily manufacturing-based, but technologically more advanced in the relevant margin.
- Direct firm-level R&D reporting is available in both settings, which is crucial because the model needs an observable proxy for the innovation decision.

Interpretation. The cross-country comparison is not a side exercise. It lets the paper ask whether China's problem is only too little innovation, or whether it is also that Chinese R&D is less productive and more distorted than in a nearby benchmark economy.

2.2 Balanced-panel sample and basic summary facts

Table I summarizes the core samples.

- China balanced panel: 123,368 firms in both 2007 and 2012; 18,140 report positive R&D in 2007.
- Private-firm China balanced panel: 117,983 firms in 2007, of which 15,828 report positive R&D.
- Taiwan balanced panel: 11,229 plants in both 1999 and 2004; 1,487 report R&D in 1999.

For China, the median firm in the balanced panel has value added of about 1.48 million USD in 2007 and 3.33 million USD in 2012. The aggregate R&D intensity is 1.86% in 2007. In Taiwan, aggregate R&D intensity is higher, around 3%.

Interpretation. The Chinese sample is much larger and the firms are much bigger on average, but Taiwan looks more R&D-intensive. That becomes important later when the paper compares estimated innovation technology across the two economies.

2.3 How the paper maps data into innovation and imitation

The main empirical proxy is the *extensive margin* of R&D.

- Firms with positive R&D expenditure are classified as **innovators**.
- Firms with zero R&D expenditure are classified as **imitators**.

The authors focus on the extensive margin because it matches the model's discrete choice, because fixed costs of running an R&D lab are important, and because the intensive margin is measured with more error.

The sample excludes the bottom 10% of the TFP distribution because those firms are extremely small and measurement error is likely to be especially severe there.

Interpretation. This is a coarse proxy, but it is tightly aligned with the model: the firm is choosing whether to pursue an innovation strategy or an imitation strategy.

2.4 Measuring TFP and output wedges

The production side follows Hsieh and Klenow (2009). Firm output is

$$Y_i(t) = A_i(t)K_i(t)^\alpha L_i(t)^{1-\alpha}. \quad (1)$$

Given monopolistic competition and wedges, the static equilibrium implies

$$\pi_i(t) \propto A_i(t)(1 - \tau_i(t))^{\eta-1}, \quad (2)$$

and

$$P_i(t)Y_i(t) \propto A_i(t)(1 - \tau_i(t))^{\eta-1}. \quad (3)$$

Combining these expressions yields the TFP formula used in the data work:

$$A_i(t) \propto \frac{(P_i(t)Y_i(t))^{\eta/(\eta-1)}}{K_i(t)^\alpha N_i(t)^{1-\alpha}}. \quad (4)$$

The paper sets $\eta = 5$ and lets the capital share α vary across industries with the industry labor share. TFP is then residualized on industry, province, and age effects.

Interpretation. A wedge lowers size and profits relative to TFP. This is why, in the data, firm size contains information about distortions over and above the firm's productivity level.

2.5 The four descriptive moments that discipline the model

Figure 2 is the empirical backbone of the paper. It contains four objects:

- Panel A: fraction of R&D firms by TFP percentile.
- Panel B: fraction of R&D firms by value-added percentile.
- Panel C: TFP growth of non-R&D firms by initial TFP percentile.
- Panel D: TFP-growth difference between R&D and non-R&D firms by initial TFP percentile.

For estimation, the authors compress these graphs into four percentile bins: 10–49, 50–79, 80–94, and 95–99, producing sixteen target moments.

Interpretation. These four panels line up almost one-for-one with the theoretical logic of the paper: selection into innovation, selection into innovation through wedges and firm size, imitation-driven convergence, and the innovation premium.

2.6 Measurement error

Measurement error is central because it affects both TFP and wedges. The paper assumes classical measurement error in value added and inputs:

$$\ln(\widehat{P_i Y_i}) = \ln(P_i Y_i) + \mu_y, \quad \ln(\widehat{K_i^\alpha N_i^{1-\alpha}}) = \ln(K_i^\alpha N_i^{1-\alpha}) + \mu_I. \quad (5)$$

Assuming firm-specific wedges are constant over 2007–2012, the covariance of five-year growth in value added and inputs identifies the variance of true value-added growth, and hence the variance of measurement error:

$$\text{cov}(\Delta \ln(P_i Y_i), \Delta \ln(K_i^\alpha N_i^{1-\alpha})) = \text{var}(\Delta \ln(P_i Y_i)). \quad (6)$$

The implied measurement error in log TFP is

$$a_i - \widehat{a}_i = \frac{\eta}{\eta - 1} \mu_y - \mu_I, \quad (7)$$

so its variance is

$$\widehat{v}_{\mu_a} = \left(\frac{\eta}{\eta - 1} \right)^2 \widehat{v}_{\mu_y} + \widehat{v}_{\mu_I}. \quad (8)$$

Interpretation. Measurement error flattens the relation between R&D and observed TFP/size but exaggerates measured convergence. The paper therefore models it explicitly rather than treating the descriptive panels as clean moments.

2.7 How wedges are parameterized in the cross section

The model imposes a simple relationship between wedges and TFP:

$$-\ln(1 - \tau_i) = b(a_i - \bar{a}) + \varepsilon_i^\tau. \quad (9)$$

The paper corrects the estimated slope b for measurement error and then uses the implied residual variance of wedges in the simulations.

Interpretation. This parameterization matters a lot because the correlation between TFP and wedges is what drives the gap between the TFP–R&D profile and the size–R&D profile. If wedges were irrelevant, Panels A and B of Figure 2 would look much more similar.

3 Models and empirical specifications (all equations + interpretation)

3.1 Static side of the model

The final good is a Dixit–Stiglitz aggregate of monopolistically competitive varieties. Given the static structure, current profits and firm size are both increasing in TFP and decreasing in the output wedge:

$$\pi_i(t) \propto A_i(t)(1 - \tau_i(t))^{\eta-1}, \quad P_i(t)Y_i(t) \propto A_i(t)(1 - \tau_i(t))^{\eta-1}. \quad (10)$$

Interpretation. The static side is standard Hsieh–Klenow. The new action is dynamic: firms use current profitability and distortions to decide whether to spend resources on future productivity growth.

3.2 Innovation and imitation without R&D costs

Firms move on a productivity ladder. A successful step adds a constant amount $\tilde{a}$ to log TFP. Low-TFP firms can improve through imitation; high-TFP firms may need to innovate.

If a firm imitates, it meets another firm at random and improves by one rung with probability $q(1 - F_a)$, where F_a is the cumulative TFP distribution up to rung a .

If a firm innovates, it draws an idiosyncratic success probability p and succeeds either directly or, if direct innovation fails, through a second-chance imitation channel with probability $\delta q(1 - F_a)$.

Without R&D costs, the firm innovates if and only if

$$p_i \geq Q(a, \tau; A) \equiv \frac{q(1 - \delta)(1 - F_a)}{1 - \delta q(1 - F_a)}. \quad (11)$$

Interpretation. The threshold falls with TFP rank a . That is the paper’s most important selection result: the farther a firm is from the frontier, the better imitation looks; the closer it is to the frontier, the more innovation becomes necessary.

3.3 Costly innovation and the role of wedges

With costly innovation, firm i ’s discounted profit is written as

$$\pi_i(t) = \frac{1}{1 + r}(1 - \tau_i(t))A_i(t)^{\eta-1} \tilde{\Pi}(t), \quad (12)$$

while the innovation cost is

$$c_i(t) = (A_i(t)^\theta \bar{A}(t)^{1-\theta})^{\eta-1} \bar{c} \tilde{\Pi}(t). \quad (13)$$

The innovation threshold becomes

$$Q(a, \tau_j; A) = \frac{q(1 - \delta)(1 - F_a)}{1 - \delta q(1 - F_a)} + \frac{e^{(1-\theta)(\eta-1)(a-\bar{a})}}{e^{\eta-1} - 1} \mathbb{E}[(1 - \tau')^{\eta-1} | \tau_j] \frac{\bar{c}(1 + r)(1 + g)^{\eta-2}}{1 - \delta q(1 - F_a)}. \quad (14)$$

Interpretation. Two effects show up immediately.

- A higher wedge lowers the future payoff from being more productive, so it discourages innovation.
- A higher TFP level still makes innovation relatively more attractive because imitation opportunities shrink near the frontier.

This is the central comparative-advantage logic of the paper.

3.4 Law of motion and the traveling-wave idea

The full model implies an integro-difference law of motion for the TFP distribution. In words,

$$A_a(t+1) - A_a(t) = \text{inflow from successful firms at } a-1 - \text{outflow from successful firms at } a. \quad (15)$$

The main theoretical result is that the economy admits a traveling-wave stationary solution,

$$A_a(t) = f(a - \nu t), \quad (16)$$

with speed $\nu > 0$ and Pareto tails on both the low- and high-TFP sides.

Interpretation. “Stationary” here does *not* mean no growth. It means that after removing a constant endogenous drift, the shape of the TFP distribution is stable and simply travels to the right over time.

3.5 Testable predictions of the theory

The paper takes the following four predictions to the data:

1. conditional on other observables, the share of R&D firms rises with TFP;
2. firms with higher wedges are less likely to invest in R&D, so conditional on TFP, larger firms are more likely to be R&D firms;
3. expected TFP growth is decreasing in initial TFP, especially among non-R&D firms;
4. the TFP-growth gap between R&D and non-R&D firms rises with TFP.

Interpretation. Figure 2 is almost a direct visual translation of these four predictions.

3.6 Calibration and SMM objective

Some parameters are calibrated externally or from auxiliary moments.

- $\eta = 5$ from the Hsieh–Song implementation.
- $\theta \approx 0.25$ from the relative R&D-cost-to-value-added ratio of low- versus high-TFP firms.
- $\tilde{a} = 0.78$ to imply a realistic average innovation cost, about 3.7% of industrial value added in the benchmark economy.

The remaining parameters are estimated by simulated method of moments:

$$\hat{\phi} = \arg \min_{\phi} h(\phi)' W h(\phi), \quad (17)$$

where $h(\phi)$ is the vector of percentage deviations between simulated and empirical moments and the benchmark weighting matrix is the identity matrix.

Interpretation. The paper does not estimate the model by fitting every micro observation. It estimates the parameters that make the model reproduce the key empirical shapes in the data.

3.7 What each parameter is pinned down by

The paper is unusually explicit about identification inside the structural model.

- q is mostly pinned down by TFP convergence among non-R&D firms (Panel C of Figure 2), after correcting for measurement error.

- δ is pinned down by how much convergence remains among R&D firms and hence by the relative slopes of Panels C and D.
- $\bar{p}$ is pinned down by the TFP-growth premium of R&D firms relative to non-R&D firms.
- $\bar{c}$ is disciplined by the overall fraction of firms choosing R&D.
- σ_{μ_a} is identified from the joint fact that measurement error flattens Panels A and B but steepens Panel C.

Interpretation. In a structural note, this subsection is the analogue of an identification section in a reduced-form paper. Each parameter is tied to one piece of the empirical geometry.

3.8 Model variants beyond the parsimonious benchmark

The benchmark model is the **PAM** (parsimonious model), which estimates five objects: q , δ , $\bar{p}$, $\bar{c}$, and σ_{μ_a} .

The **FLM** (flexible model) adds idiosyncratic innovation-cost heterogeneity:

$$c_i(t) \propto \left[\bar{c} - \exp\left(\xi_i(t) - \frac{\sigma_c^2}{2}\right) + 1 \right] (A_i(t)^\theta \bar{A}(t)^{1-\theta})^{\eta-1}. \quad (18)$$

The **IPM** (industrial policy model) lets innovation wedges depend on TFP rank:

$$c_i(t) \propto \left[\bar{c} - \exp\left(\xi_i(t) - \frac{\sigma_c^2}{2}\right) + 1 + c_a(G(a_i) - 1/2) \right] (A_i(t)^\theta \bar{A}(t)^{1-\theta})^{\eta-1}. \quad (19)$$

A positive c_a means low-TFP firms are subsidized relative to high-TFP firms.

The **FRM** (fake R&D model) adds a share Υ of firms that can claim subsidies without actually undertaking productive innovation effort.

Interpretation. These extensions are not cosmetic. They ask whether the benchmark model fails because innovation costs vary across firms, because policy explicitly tilts subsidies toward some firms, or because some reported R&D is not real innovation at all.

4 Identification and instruments (what makes the estimates credible)

4.1 What identifies the main estimate

The paper has no single IV-like exclusion restriction. Credibility comes from matching a very tight bundle of facts simultaneously:

- selection into R&D by TFP,
- selection into R&D by size,
- TFP convergence among non-R&D firms,

- the conditional TFP-growth premium of R&D firms,
- and auxiliary moments on measurement error.

Interpretation. A model that gets only one of these right is not enough. The content is in matching the joint pattern, especially the fact that size predicts R&D more strongly than TFP in the data.

4.2 Why the measurement-error strategy matters

A big concern in Hsieh–Klenow-style exercises is that measured TFP and measured wedges are mechanically contaminated by the same accounting noise. The paper addresses this head-on by using five-year growth covariances and the assumption that wedges are constant over 2007–2012.

In the benchmark PAM, the estimated measurement error accounts for about 30% of the variance of log TFP and about 92% of the variance of TFP growth.

Interpretation. This is why the authors resist reading Figure 2 naively. Much of the apparent convergence in raw TFP growth is accounting noise, not real technological catch-up.

4.3 Why the wedge parameterization is informative

The key reduced-form fact is that the R&D–size profile is much steeper than the R&D–TFP profile. In the model, that can happen only if wedges matter, because size is the combination of TFP and wedges.

Interpretation. This is an elegant identification device. If R&D depended only on TFP, Panels A and B of Figure 2 would be much more alike. Their difference is what pins down the role of distortions.

4.4 What the extensions are identified from

The extra parameters in the flexible models are disciplined by very specific failures of the benchmark.

- σ_c is identified because adding idiosyncratic innovation-cost dispersion flattens the overly steep Panels A and B of the PAM.
- c_a is identified because the IPM can flatten Panel A without over-flattening Panel B, which the FLM tends to do.
- Υ in the fake-R&D model is identified from the possibility that many firms report R&D but do not generate the productivity patterns one would expect from truthful innovation.

Interpretation. Each extension is motivated by a particular residual mismatch. This is good structural practice: do not add parameters abstractly; add them to solve a concrete empirical failure.

4.5 Why the Taiwan comparison is useful

Taiwan is estimated with the same structure and the same TFP step size. This means the Chinese and Taiwanese estimates are directly comparable.

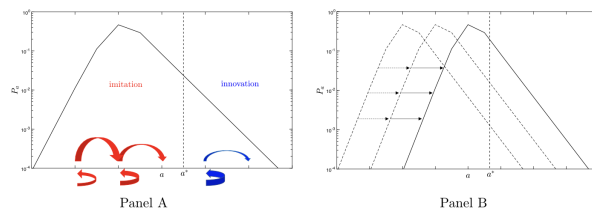


FIGURE 1.—Equilibrium dynamics with a stationary distribution. *Note:* Panel A displays the threshold a^* and the stationary TFP distribution. Panel B plots a traveling wave.

Interpretation. The Taiwan exercise is doing two jobs at once: it is a robustness check for the model class and a decomposition tool for thinking about whether China’s lower growth comes from weaker innovation technology, slower diffusion, or greater misallocation.

4.6 What the paper can and cannot distinguish

- The paper infers wedges from the Hsieh–Klenow methodology, so its conclusions inherit the strengths and limits of that framework.
- The theory abstracts from firm entry and exit and focuses on incumbents in a balanced panel.
- The extensive-margin R&D proxy is deliberately coarse; the intensive margin and project size are only partly explored in robustness work.
- The IPM and FRM are not always sharply distinguishable in the data. The estimation can have local minima that fit similarly well.

Interpretation. The paper is strongest on the broad message that misallocation distorts the innovation–imitation margin. It is less decisive on whether the distortion comes mainly from badly targeted industrial policy, fake R&D reporting, or some mixture of both.

5 Tables and figures

The paper’s evidence is organized very cleanly: first show the theory’s geometry, then the China moments, then the reduced-form regressions, then the structural fit, then the untargeted moments, then Taiwan, and finally the counterfactuals.

5.1 Figure 1: the theoretical traveling wave

Read:

- Figure 1 illustrates the stationary traveling-wave equilibrium. In Panel A, low-TFP firms lie on the imitation side and high-TFP firms lie on the innovation side.
- Panel B shows the whole TFP distribution drifting to the right over time without changing its shape once one removes the constant endogenous growth trend.

Economic message. This figure is the paper’s core picture. Growth comes from a moving distribution, not from a representative firm. That makes the assignment of firms to imitation

versus innovation quantitatively central.

5.2 Table I and Figure 2: the China data line up with the model's four predictions

TABLE I
SUMMARY STATISTICS.

Year	Number of Firms	Number of R&D Firms	Median Value Added (Million USD)	Mean Value Added (Million USD)	Median R&D Intensity (%)	Aggregate R&D Intensity (%)
Balanced Panel of Chinese Firms						
2007	123,368	18,140	1.48	5.81	1.73	1.86
2012	123,368	N.A.	3.33	11.45	N.A.	N.A.
Private Chinese Firms in the Balanced Panel						
2007	117,983	15,828	1.43	4.67	1.65	1.54
2012	117,771	N.A.	3.26	9.57	N.A.	N.A.
Balanced Panel of Taiwanese Firms						
1999	11,229	1,487	0.16	2.91	8.50	3.14
2004	11,229	1,144	0.17	4.78	6.42	2.93

Note: R&D intensity is the ratio of R&D expenditure to value added. Median R&D intensity is the median R&D intensity among firms performing some R&D. Aggregate R&D intensity is the ratio of aggregate R&D expenditure to aggregate value added for all firms. Missing information is due to the lack of R&D data in 2012. Firm-level value added is not available after 2008. We use the average industry-specific ratio of value added to sales between 2004 and 2007 to impute firm value added based on observed firm sales after 2008.

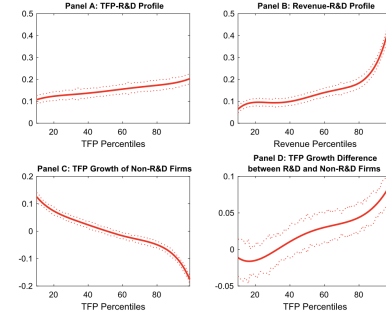


FIGURE 2.—Chinese firms in the balanced panel 2007–2012. *Note:* The X-axis in Panels A, C, and D is the 2007 TFP percentile. The X-axis in Panel B is the 2007 value added percentile. The solid lines in Panels A and B plot the 2007 fraction of R&D firms in each TFP and value added percentile, respectively. The solid line in Panel C plots the median annualized 2007–2012 TFP growth among nonR&D firms in each TFP percentile. The solid line in Panel D plots the difference between the median 2007–2012 TFP growth R&D and nonR&D firms within each percentile. A firm's TFP growth is the residual of the regression of TFP growth on industry, age, and province fixed effects. All the solid lines are smoothed by a fifth-order polynomial. The dotted lines plot the 95% confidence intervals by bootstrap.

Read:

- Table I shows that Chinese firms are larger than Taiwanese firms and that only about 15% of firms report positive R&D in the balanced 2007 China sample.
- Figure 2, Panel A shows that the probability of doing R&D rises with TFP.
- Figure 2, Panel B shows that the probability of doing R&D rises even more steeply with firm size.
- Figure 2, Panel C shows strong TFP convergence among non-R&D firms.
- Figure 2, Panel D shows that R&D firms grow faster than non-R&D firms, especially higher up in the TFP distribution.

Economic message. This is the most important descriptive page in the paper. The fact that size predicts R&D more strongly than TFP is exactly the empirical footprint of wedges distorting the innovation decision.

5.3 Table II: reduced-form regressions say the same thing in regression form

Read:

- In Panel A, the probability of doing R&D is positively related to log TFP and negatively related to the output wedge. Once the wedge is included, the TFP coefficient rises from 0.062 to 0.368 and the wedge coefficient is about -0.410 .
- In Panel B, initial log TFP enters negatively in the TFP-growth regression with a coefficient of about -0.062 , while the R&D dummy enters positively with a coefficient around 0.036 to 0.037.

TABLE II
BALANCED PANEL OF CHINESE FIRMS, 2007–2012.

PANEL A Dependent variable: R&D decision in 2007				
	(1) R&D _{<i>it</i>}	(2) R&D _{<i>it</i>}	(3) R&D _{<i>it</i>}	(4) R&D _{<i>it</i>}
log(TFP)	0.062 (0.0074)	0.368 (0.0284)	0.343 (0.0259)	0.305 (0.0232)
wedge		-0.410 (0.0357)	-0.378 (0.0323)	-0.332 (0.0296)
export _{<i>it</i>}			0.053 (0.0134)	0.054 (0.0132)
SOE _{<i>it</i>}				0.205 (0.0232)
R-squared	0.143	0.208	0.211	0.224

PANEL B Dependent variable: TFP growth					
	(1) TFP growth	(2) TFP growth	(3) TFP growth	(4) TFP growth	(5) TFP growth
log(TFP)	-0.062 (0.0035)	-0.062 (0.0036)	-0.062 (0.0035)	-0.062 (0.0036)	-0.062 (0.0036)
R&D _{<i>it</i>}	0.036 (0.0042)	0.037 (0.0040)	0.034 (0.0035)		
export _{<i>it</i>}		-0.006 (0.0038)	-0.006 (0.0035)	-0.006 (0.0037)	-0.007 (0.0035)
SOE _{<i>it</i>}			0.029 (0.0115)		0.029 (0.0113)
R&D intensity _{<i>it</i>}				0.044 (0.0060)	0.041 (0.0058)
R&D intensity _{<i>it</i>} _m				0.042 (0.0069)	0.038 (0.0056)
R&D intensity _{<i>it</i>} _y				0.025 (0.0035)	0.023 (0.0033)
R-squared	0.122	0.122	0.123	0.122	0.123

Note: Panel A: the dependent variable R&D_{*it*} is a dummy variable switching on for firms that report positive R&D expenditure. Panel B: the dependent variable is annualized TFP growth 2007–2012. Explanatory variables: log(TFP) is the logarithm of TFP; Wedge indicates the firm's output wedge $-\log(1 - \tau_i)$ and is calculated as described in Section 2; export_{*it*} is a dummy variable for exporters; SOE_{*it*} is a dummy variable for state-owned firms; R&D intensity_{*it*} is a dummy variable for high R&D intensity switching on if the firm R&D expenditure over sales is in the 67th percentile and above (among all R&D firms); R&D intensity_{*it*}_m is the analogue dummy for medium R&D intensity (between 33rd and 66th percentiles); R&D intensity_{*it*}_y is the analogue dummy for low R&D intensity (below the 33rd percentile.) All the explanatory variables are from 2007. Standard errors are reported in parentheses. The number of observations is 109,799. Observations are weighted by employment and standard errors are clustered by industry. All regressions include industry, age, and province fixed effects. We drop firms with TFP in the bottom 10 percentiles.

- When the intensive margin is split into terciles, all three R&D-intensity dummies are positive, and the high-intensity group has the largest coefficient (about 0.044).

Economic message. Table II matters because it shows that the graphical patterns are not artifacts of smoothing. The two reduced-form relations the structural model most needs to match—selection into R&D and higher future TFP growth for R&D firms—both appear clearly in regression form.

5.4 Figure 3 and Table V: Taiwan looks similar, but innovation works better there

Read:

- Figure 3 shows that Taiwan has the same qualitative four-panel patterns as China.
- The R&D–TFP profile is steeper in Taiwan: moving from the 60th to the 99th percentile, the R&D share rises from roughly 10% to above 35%.
- Table V shows that both imitation and innovation are stronger in Taiwan. In the PAM, $q = 0.286$ in Taiwan versus 0.175 in China, and $\bar{p} = 0.184$ versus 0.096.
- Innovation is also costlier in Taiwan ($\bar{c} = 3.301$ in the PAM versus 1.627 in China), but the productivity improvement dominates.

Economic message. The Taiwan comparison says that China's problem is not simply that firms do too little R&D. A large part of the problem is that R&D is less productive and diffusion is slower than in a nearby manufacturing economy.

TABLE V
ESTIMATION FOR TAIWAN, 1999–2004.

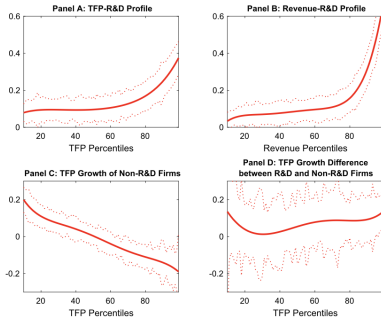


FIGURE 3.—Taiwanese firms in the balanced panel 1999–2004. *Note:* This figure is the analogue of Figure 2 for the sample of Taiwanese firms.

	PAM	FLM	IPM
Imitation prob. q	0.286 (0.086)	0.501 (0.129)	0.371 (0.120)
Second chance δ	0.027 (0.023)	0.002 (0.058)	0.001 (0.047)
Innov. prod. $\bar{p}$	0.184 (0.026)	0.207 (0.031)	0.206 (0.032)
Innov. cost $\bar{c}$	3.301 (0.475)	4.669 (0.717)	5.247 (0.809)
Std.dev. m.e. $\sigma_{\mu a}$	0.722 (0.074)	0.538 (0.073)	0.622 (0.112)
Std.dev. innov. subs. σ_c		1.371 (0.146)	1.378 (0.108)
Policy inter. c_a			−2.385 (0.792)
J-statistic	0.861	0.570	0.503

Note: Estimated parameters of various models using the Taiwanese 1999–2004 balanced panel sample. Bootstrapped standard errors in parentheses.

5.5 Table III and Figures 4–5: the structural models fit the China moments, and the extensions matter

TABLE III ESTIMATION, CHINESE FIRM BALANCED PANEL 2007–2012.						
	(1)	(2)	(3)	(4)	(5)	(6)
	PAM	FLM	IPM	FRM	FRM	IPM
Imitation prob. q	0.175 (0.031)	0.271 (0.019)	0.361 (0.019)	0.275 (0.051)	0.294 (0.058)	0.546 (0.052)
Second chance δ	0.008 (0.011)	0.020 (0.021)	0.001 (0.027)	0.019 (0.026)	0.141 (0.106)	0.001 (0.080)
Innov. prod. $\bar{p}$	0.096 (0.008)	0.114 (0.006)	0.113 (0.006)	0.237 (0.016)	0.107 (0.015)	0.111 (0.010)
Innov. cost $\bar{c}$	1.627 (0.136)	3.374 (0.174)	2.518 (0.177)	10.486 (1.363)	3.601 (0.448)	9.385 (2.403)
Std.dev. m.e. $\sigma_{\mu a}$	0.549 (0.014)	0.472 (0.008)	0.431 (0.005)	0.459 (0.025)	0.476 (0.022)	0.391 (0.011)
Std.dev. innov. subs. σ_c		1.243 (0.038)	1.092 (0.037)	0.011 (0.036)	1.969 (0.213)	
Policy inter. c_a		1.888 (0.159)	−10.46 (1.042)		2.499 (0.356)	
Fake share Υ			0.099 (0.005)			
J-statistic	1.518	0.507	0.368	0.362	2.690	0.516

Note: The table shows the estimated parameters for the different models discussed in the text. Columns (1)–(4) are for the Perseus (PAM), Flexible (FLM), Industrial Policy (IPM), and Fake R&D (FRM) model, respectively. Columns (5)–(6) present the results for the models estimated on moments applying a more stringent definition of R&D. Bootstrapped standard errors in parentheses.

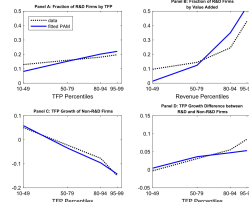


FIGURE 4.—Perseus model (PAM). *Note:* The figure shows the fit of the PAM. It plots the moments predicted by the model against their empirical counterparts for China 2007–2012. The X-axis in Panels A, C, and D is the first period TFP percentiles defined on four intervals: the 10th to 49th, the 50th to 79th, the 80th to 94th, and the 95th to 99th. The X-axis in Panel B is the first period value added percentiles defined on the same four intervals. The solid lines in Panels A and B plot the first-period fraction of R&D firms in each TFP and value added interval, respectively. Panel C plots the median annualized TFP growth among nonR&D firms in each TFP interval in the data against the corresponding expected growth rate for firms in the model. Panel D plots the difference between the median TFP growth between R&D and nonR&D firms in the data against the corresponding difference in expected growth in the model. A firm's TFP growth is the residual of the regression of TFP growth on industry, age, and province fixed effects.

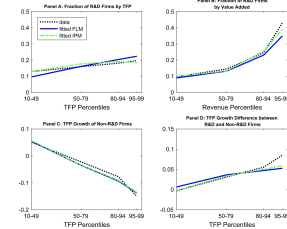


FIGURE 5.—Flexible (FLM) and Industrial Policy (IPM) models. *Note:* The figure shows the fit of the FLM and IPM. See Figure 4 for additional information.

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- Table III reports the main parameter estimates. In the PAM, $q = 0.175$, $\delta = 0.008$, $\bar{p} = 0.096$, $\bar{c} = 1.627$, and $\sigma_{\mu a} = 0.549$.
- In the PAM, the average innovation success probability is only about $\bar{p}/2 \approx 4.8\%$.
- The J-statistic falls from 1.518 in the PAM to 0.507 in the FLM and to 0.368 in the IPM, so allowing heterogeneous innovation costs substantially improves fit.
- In the IPM, the policy parameter c_a is positive (1.888), implying that low-TFP firms are effectively subsidized relative to high-TFP firms.
- In the FRM, the estimated fake-R&D share is about $\Upsilon = 0.099$, and the innovation productivity parameter jumps to $\bar{p} = 0.237$.

Economic message. The benchmark already gets the big picture right, but the flexible models teach the deeper lesson: the Chinese data look like a world where innovation costs are heterogeneous and where policy may be tilting the R&D decision toward relatively low-TFP firms.

5.6 Table IV and Figures 6–7: the untargeted moments mostly support the theory

TABLE IV
INDIRECT INFERENCE, BALANCED PANEL OF CHINESE FIRMS, 2007–2012.

PANEL A Dependent Variable: R&D decision in 2007					
	(1)	(2)	(3)	(4)	(5)
	Data	PAM	FLM	IPM	FRM
log(TFP)	0.368 (0.0284)	0.712 (0.0145)	0.287 (0.0167)	0.298 (0.0181)	0.322 (0.0167)
wedge	-0.410 (0.0337)	-0.824 (0.0177)	-0.274 (0.0202)	-0.327 (0.0214)	-0.357 (0.0203)

PANEL B Dependent Variable: TFP Growth					
	(1)	(2)	(3)	(4)	(5)
	Data	PAM	FLM	IPM	FRM
log(TFP)	-0.062 (0.0033)	-0.084 (0.0002)	-0.104 (0.0003)	-0.112 (0.0003)	-0.081 (0.0004)
R&D _i	0.036 (0.0042)	0.034 (0.0005)	0.033 (0.0005)	0.028 (0.0005)	0.053 (0.0008)

Note: The first column reports the regression results from the data. All regressions include industry, age, and province fixed effects. Columns (2)–(5) report the results from the simulated data. See Table II for variable definitions.

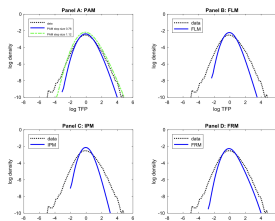


FIGURE 6.—TFP distribution. Note: The dotted lines represent the log density of log TFP in the data. The solid and dashed lines are the steady-state log density of log TFP for various models. Panel A shows two versions of the PAM—the benchmark model and the one with smaller step size. Panel B: FLM, Panel C: IPM, and Panel D: FRM.

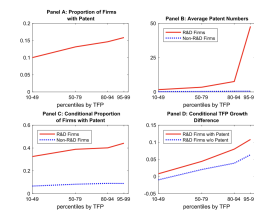


FIGURE 7.—Patents of Chinese firms, balanced panel 2007–2012. Note: All patents in the figure refer to invention patents applied for during 2007–2012. We group all firms in the 2007–2012 balanced panel into percentiles by their initial TFP. Panel A plots the proportion of firms with patents in each percentile. Panel B plots the average number of patents among R&D firms (solid line) and among nonR&D firms (dotted line) in each percentile. Panel C plots the proportion of R&D firms with one or more patents (solid line) and the proportion for nonR&D firms (dotted line). The solid line in Panel D plots the TFP growth difference between R&D firms with patents and nonR&D firms. The dotted line plots the TFP growth difference between R&D firms without patents and nonR&D firms.

Read:

- Table IV runs indirect-inference regressions. The simulated models reproduce the main data facts: R&D rises with log TFP and falls with wedges; TFP growth falls with initial TFP and is higher for R&D firms.
- Figure 6 compares the model-implied stationary TFP distribution to the data. The PAM captures about 72% of the empirical TFP variance; the heterogeneous-cost models capture about 44% to 62%.
- Figure 7 shows that patents are concentrated among R&D firms, and that the TFP-growth premium is highest for R&D firms that actually patent.

Economic message. This is reassuring because these objects were not all directly targeted. The model does not perfectly match the full TFP dispersion, but it gets the qualitative shape, the convergence logic, and the link between R&D and realized innovation outcomes broadly right.

5.7 Figure 8, Table VI, Figures 9 and 10: the counterfactuals are the pay-off to the whole exercise

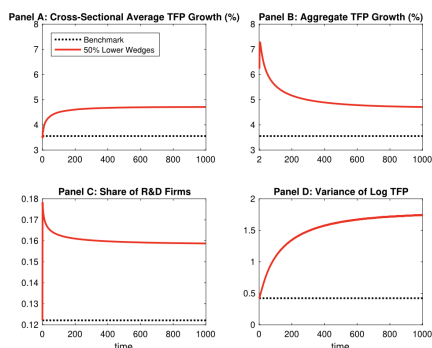


FIGURE 8.—Transition after lowering wedges. Note: The graph displays the transition following a 50% reduction in the variance of wedges. Panel A plots the growth rate of cross-sectional average TFP. Panel B plots the growth rate of aggregate TFP. Panel C plots the share of R&D firms, and Panel D plots the cross-sectional variance of log TFP.

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TABLE VI
COUNTERFACTUALS, PARSIMONIOUS MODEL.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	PAM estim. model	50% lower output wedges	Taiwan's q	Taiwan's $\bar{p}$ and $\bar{c}$	Taiwan's $\bar{p}$, $\bar{c}$, and q	Increase $\bar{c}$ so share R&D firms is 5%	Decrease $\bar{c}$ so share R&D firms is 20%	All firms do R&D
Fraction of R&D firms (%)	12.2	15.8	10.7	14.1	12.2	5	20	100
Steady-state TFP growth (%)	3.56	4.70	4.49	4.92	6.03	2.41	4.42	3.80

Note: The table reports statistics for the counterfactual experiments for the PAM discussed in the text. Column (1) reports the predicted moments of the estimated PAM for comparison.

- Figure 8 studies a 50% reduction in the variance of wedges. Aggregate TFP growth jumps up and then declines gradually toward a higher steady state; the share of R&D firms jumps from roughly 12% to nearly 18% on impact and settles around 16%; TFP dispersion rises strongly over time.
- Table VI summarizes the steady-state counterfactuals in the PAM. Baseline growth is 3.56% with 12.2% of firms doing R&D. Cutting misallocation by half raises growth to 4.70% and the R&D share to 15.8%.
- If China had Taiwan’s diffusion parameter q , growth would rise to 4.49% even though the R&D share would fall to 10.7%.
- If China had Taiwan’s full innovation technology (q , $\bar{p}$, and $\bar{c}$), steady-state TFP growth would rise all the way to 6.03% with essentially the same R&D share as in the benchmark.
- Moderate subsidies help, but too much subsidy hurts. A policy that drives the R&D share to 20% raises growth to 4.42%, but a policy that pushes all firms into innovation lowers growth to 3.80%.
- Figure 9 shows why lower wedges are so powerful: R&D becomes concentrated in high-TFP and high-value-added firms, which is exactly where the model says innovation should occur.
- Figure 10 shows the dynamic catch-up effect in the extension with international spillovers. The TFP gap relative to the frontier shrinks sharply on impact and then keeps falling during transition.

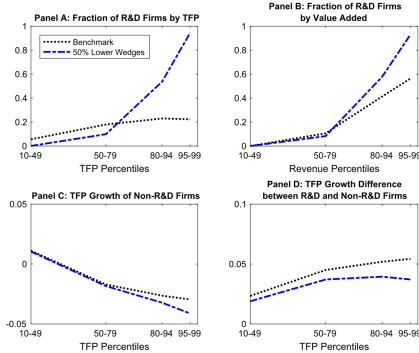


FIGURE 9.—Steady-state moments in the counterfactual model. *Note:* The dashed lines display the moments in steady state for the counterfactual experiment where we reduce the variance of wedges by 50%. The dotted line displays the moments predicted by the benchmark estimated PAM.

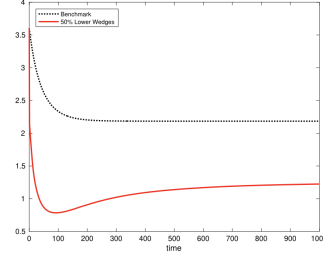


FIGURE 10.—TFP Gap during transition with international spillovers. *Note:* Evolution over time of the ratio between the TFP of the frontier economy and that of the benchmark economy (China) in the model with international spillovers. The dotted and solid lines display simulated transitions in the estimated model and in the counterfactual where output wedges are reduced by 50%, respectively.

Economic message. These are the paper’s most policy-relevant results. Misallocation hurts not only today’s static efficiency but tomorrow’s growth rate because it puts innovation in the wrong hands. And innovation subsidies are not automatically good; when imitation is valuable, indiscriminate pro-innovation policy can backfire.

6 Short summary

- The paper studies how misallocation affects the assignment of firms between innovation and imitation, and hence affects TFP growth.
- It estimates a dynamic model of Chinese manufacturing using moments on R&D participation, TFP, size, and subsequent TFP growth.

- In the data, R&D is more common among high-TFP firms and especially among larger firms, and R&D firms grow faster conditional on TFP.
- The benchmark model fits these moments well and predicts aggregate TFP growth close to the Chinese data.
- Allowing for heterogeneous innovation costs and policy wedges substantially improves fit and suggests that Chinese R&D is misallocated.
- Taiwan has both more productive innovation and faster diffusion than China.
- Reducing misallocation raises long-run growth and shifts innovation toward high-TFP firms.
- Moderate R&D subsidies help, but extreme subsidies can reduce growth because imitation is also an important engine of catch-up.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that the efficiency of growth does not depend only on how much firms spend on R&D, but on whether the right firms innovate and the right firms imitate. In the Chinese data, wedges materially distort that assignment. As a result, misallocation has dynamic effects on TFP growth over and above the static allocative losses emphasized in earlier work.

7.2 Economic intuition

A firm's distance to the frontier determines whether its best growth opportunity comes from imitation or innovation. Low-productivity firms have a lot to gain from learning from better firms; high-productivity firms have to invent. Distortions break this natural sorting. When low-TFP firms are pushed into innovation and high-TFP firms are discouraged from it, the economy loses on both fronts: it gets too little useful innovation and too little useful imitation.

7.3 Takeaway

The big takeaway is that innovation policy cannot be judged only by how much extra R&D it creates. In a world where imitation is also a growth engine, indiscriminate pro-innovation policy can be wasteful or even harmful. The real objective is not *more* innovation by itself, but a better assignment of innovative effort across firms.